

1. Introduction to C

- History of C
 - Where C is used
 - Structure of a C program
 - Compiler and execution process
 - First program
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2. Variables and Data Types

- Variables
 - Constants
 - int
 - float
 - double
 - char
 - bool
 - Type conversion
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3. Operators

- Arithmetic operators
 - Logical operators
 - Comparison operators
 - Assignment operators
 - Bitwise operators
 - Increment / decrement
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4. Input and Output

- printf
 - scanf
 - Formatting output
 - Reading user input
-

5. Conditions

- if
 - else
 - else if
 - switch
 - Nested conditions
-

6. Loops

- for
 - while
 - do while
 - break
 - continue
 - Nested loops
-

7. Functions

- Function declaration
 - Function definition
 - Parameters
 - Return values
 - Scope
 - Recursion
-

8. Arrays

- One-dimensional arrays
 - Two-dimensional arrays
 - Array traversal
 - Array algorithms
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9. Strings

- Character arrays
- String functions
- strlen

- strcpy
 - strcmp
 - strcat
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10. Pointers

- Memory addresses
 - Pointer declaration
 - Dereferencing
 - Pointer arithmetic
 - Pointers and arrays
 - Pointers and functions
 - pass by reference
 - pass by value
 - Null *pointer
 - Void pointer
 - const pointer, pointer to const
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11. Structures and Unions

- struct
 - typedef
 - Nested structures
 - union
 - Dynamically growing vector
 - Struct pointers Linked list Node
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12. Dynamic Memory Allocation

- malloc
 - calloc
 - realloc
 - free
 - Heap vs stack
 - program memory structure
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13. File Handling

- fopen
 - fclose
 - fprintf
 - fscanf
 - fread
 - fwrite
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14. Preprocessor

- #include
 - #define
 - Macros
 - Header files
 - Conditional compilation
 - linking
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15. Advanced C

- Function pointers
 - Bit fields
 - Memory management
 - Volatile
 - Static
 - Extern
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16. Data Structures in C

- Linked lists
 - Stacks
 - Queues
 - Trees
 - Hash tables
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17. Algorithms

- Searching algorithms
- Sorting algorithms
- Recursion algorithms

- Time complexity basics

digital logic design

Number Systems and Codes

Boolean Algebra

Karnaugh Maps and Function Simplification

Combinational Logic Design Principles

Combinational Logic Design with MSI Circuits

Latches and Flip-Flops

Counters and Registers

Clock-Driven Sequential Circuits

Event-Driven Circuits

Instrumentation and Interfacing

Programmable Logic Devices

Arithmetic Circuits

Fault Diagnosis and Testing

javascript engineer

1. JavaScript from low level perspective

2. JavaScript Internals

3. Functions In Memory

4. Variable Lookup and Scope

5. Variable Lookup and Closures

6. Functional Programming

7. Lexical Structure Of JS

8. Objects from High Level

9. V8 engine optimizations

10. Philosophy of Object Oriented :
Design

11. Polymorphism and Context
Invocation Details

12. Prototypes and [[Prototypes]]

13. Constructor Functions and
Inheritance Internals

14. Abstraction and Meta
Programming

15. SOLID and object oriented design

16. Creational Design Patterns

17. Structural Design Patterns

18. Behavioral Design Patterns

- 19. Process, Thread, Async
- 20. Deep Dive Async

Data Structure and Algorithms

- 1. Introduction to Algorithms and Data Structure
- 2-3. Solving Leetcode Problems(Two Pointers and other : problem solving techniques.....)
- 4. Insertion Sort and Counting Sort
- 5. Quick Sort and Merge Sort
- 6.Heaps and Priority
- Queues(Applying Sliding window, Quick Select)
- 7. Linked Lists
- 8. Problem Solving (SLL, QS, SW)
- 9. Single Linked List
- 10. Double Linked List
- 11. Hash Tables
- 12.Binary Search Trees
- 13. Binary Search Trees(Delete,DFS,BFS)
- 14. Solving LeetCode(BFS,DFS)
- 15. AVL Tree Introduction
- 16.AVL Tree implementation Details
- 17. Red Black Trees (insertion)
- 18. Red Black Tree (Deletion)
- 19.RB Trees (Delete Fixup)
- 20.Graphs
- 21.Kahn's Algorithm
(Topological Sorting)
- 20. Dijkstra and Bellman
Ford (Shortest Path)
- 21. Prim and Kruskal (minimum spanning trees)
- 22. Greedy approach
- 23-24 Dynamic Programming